

Navodit Maheshwari

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Education

M.S. Natural Language Processing, UC Santa Cruz

Sept 2025–Dec 2026

Relevant Coursework: Advanced Deep Learning, NLP, Conversational Agents

B.S. Computer Science, UC Santa Cruz

Sept 2022–June 2025

Relevant Coursework: Machine Learning, AI, Computer Systems Design, Probability Theory

Skills

Programming: Python, SQL, Bash, Java, C/C++, R, Typescript

AI/ML: PyTorch, TensorFlow, Scikit-learn, Hugging Face, XGBoost, LightGBM

LLM Systems: RAG, Agentic Workflows, Tool Calling, Vector Search, LLM Evaluation, Fine-Tuning

Infrastructure: Docker, MLflow, Git, Linux, Spark, Ray, Databricks, AWS, GCP, Firebase Firestore

Data & Visualization: Pandas, NumPy, Matplotlib, Model Benchmarking

Experience

Graduate Researcher, Wang Research Lab

Feb 2026 – Present

- Researching LLM safety with a focus on representation steering, analyzing refusal behaviors encoded in residual streams.
- Applying weight-level interventions to open-weight LLMs via difference-of-means refusal direction orthogonalization.
- Evaluating steering effects across agentic harm and prompt injection benchmarks using Inspect AI and LlamaGuard3.
- Co-authored EMNLP '26 submission showing refusal collapse from 29-85% to 0-3% across 4 model families.

ML Engineering Intern, UCSC University Business Services

Jan 2026–Present

- Collaborating in an agile team to develop a multi-agent LLM assistant streamlining 3,000+ annual UC procurement requests.
- Engineering a domain-adapted RAG pipeline with Firestore vector search over 32 policy documents for compliance retrieval.
- Curating 1,000 historical procurement records to fine-tune and evaluate conversational LLMs for policy guidance.
- Implementing Firebase authentication with role-based access control for clearance-level policy retrieval.

Tutor & Grader, Baskin School of Engineering

Sept 2024–Jun 2026

- Hosted tutoring sessions for 200+ students, adapting explanations to individual learning styles and skill levels.
- Contributed to classes such as Natural Language Processing, Computer Architecture, and Electrical Engineering.
- Supported TAs in building an efficient grading pipeline for assignments to ensure prompt feedback to students.
- Collaborated with faculty to develop and integrate a PyTorch unit into the NLP curriculum to better prepare students for industry ML roles.

Projects

Autonomous Minecraft AI Agent Framework - [GitHub](#)

Jun 2026

- Designed a stateful autonomous LLM agent for long-horizon Minecraft survival tasks, achieving 100% success on diamond tool acquisition with 33% fewer iterations than a published baseline.
- Engineered a persistent memory framework combining RAG, knowledge graphs, and episodic reflection to accumulate reusable skills across tasks.
- Integrated custom Mineflayer tools through the Model Context Protocol (MCP), enabling structured tool calling for low-level game interactions.
- Built a self-correction pipeline with world-state verification and autonomous replanning for failed execution paths.

AI Fitness Agent - [GitHub](#)

Mar 2026

- Implemented a multi-agent fitness assistant using GPT-OSS, Qwen2.5, and Gemma3 to coordinate workout planning, scheduling, and multimodal exercise assistance.
- Built RAG pipelines integrating calendar events, health metrics, and visual exercise recognition.
- Benchmarked open-source reasoning and vision models across hallucination rate, recall, and recommendation consistency metrics.

X-Ray Fracture Detection - [GitHub](#)

Mar 2026

- Collaborated on a medical-image captioning pipeline using CLIP, BLIP, and BiomedCLIP over 60K MIMIC-CXR studies.
- Implemented mixed-precision training and hard negative mining to stabilize contrastive learning on an unbalanced dataset.
- Developed retrieval-based evaluation pipelines using cosine-similarity ranking and METEOR scoring to benchmark fine-tuned models against ClinicalBLIP baselines.
- Achieved a $3.4\times$ METEOR score improvement while maintaining efficient GPU memory usage under constrained hardware resources.